GlucoTwin

A virtual patient workspace for glucose forecasting

Historical clinical context + wearable sensor history

Research prototype | 11 October 2026

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Problem and healthcare use case

A clinician needs to see how recent glucose changes relate to a patient's historical clinical context.

GlucoTwin links the profile to visible sensor history and fixed future forecasts.

The research objective is advance prediction of a new sustained high-glucose episode.

Linked historical and sensor data

CGMacros historical input: age, recorded sex, BMI, baseline HbA1c and fasting glucose.

Dynamic input: CGM glucose and heart rate, linked to the same patient.

Retrospective replay; CGM samples are reconstructed and clinical availability is assumed.

Fusion inside the model

Both streams enter the same fitted feature vector; the profile is more than display context.

CGMacros uses 5 historical fields, 22 CGM features and 9 heart-rate features.

Imputation and scaling use fitting patients only. Fusion's predictive advantage remains uncertain.

Runtime and research architecture

Historical profile + available sensor history -> causal features -> saved fused Ridge -> local Python API -> browser replay.

Separate saved XGBoost path -> uncalibrated onset score and frozen cutoff -> research decision ledger with delayed outcomes.

Patient replay uses saved models and the fold that excluded this patient. Research decisions remain separate from clinical alerts.

Clinician interaction demonstration

Inspect historical profile, sensor coverage and visible glucose history.

Advance the replay; forecasts stay fixed when issued. Outcomes appear only when available.

Compare model error with persistence, switch patients and examine missing observations.

CGMacros event research: fixed score/cutoff/decision; outcome hidden until issue time +45 minutes.

Advance-event definition

New sustained high onset: 4 consecutive five-minute readings above 180 mg/dL, after 7 readings at or below 180.

Prediction window: onset 5-30 minutes ahead. Features contain only past information.

27 development patients; 3 outer folds separate fitting, cutoff setting and evaluation. Six reserved patients remain unscored.

Measured event prediction

Metric	Fused XGBoost v2
Episodes detected	229 / 336 (68.2%)
Conservative precision	52.8%
False alarms / eligible day	0.98
Median detected-event lead	10 minutes

Development results. Cutoff-setting folds were not consistently successful; patient intervals cross gate limits.

Numerical forecast results

Cohort / horizon	Ridge MAE	Persistence MAE
CGMacros / 60 min	19.00	19.99
RBG / 30 min	17.59	20.71
RBG / 60 min	28.89	32.53

Units: mg/dL. Separate development cohorts and timing protocols. RBG profile-fused values match the original CGM-only benchmark after rounding.

Current model limitations

Only 36.6% of evaluable high episodes received at least ten minutes warning. Cutoff-setting folds did not consistently pass.

Low-glucose numerical errors remain a weakness. Overall MAE is not a clinical safety guarantee.

CGMacros samples are reconstructed; historical profile availability is assumed. A useful clinical-fusion advantage is not established.

No clinical validation or reserved-test performance. Risk percentages, operational alerts and treatment advice remain disabled.

Reproducibility and remaining work

Python + scikit-learn + XGBoost; HTML/CSS/JavaScript browser. Measured studies ran on local CPU.

The repository contains the active application, model preparation code, essential documentation and current-model checks. Records and model binaries stay local.

Next: freeze a final model/cutoff policy before using reserved data. Research replay does not establish clinical reliability.

Original code: MIT License. Dataset terms remain separate. Video preparation is deferred.